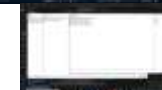
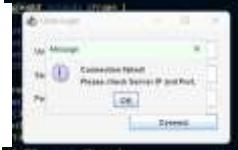
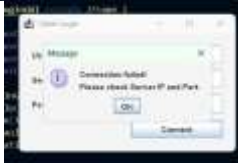


TC ID	Test Case	Description	Input	Expected Result	Status
TC01	Server start	Verify that the server starts successfully and listens for client connections	1.Open the project in VS Code 2.Open terminal 3. Type cd code 4.Type javac client/*.java javac gui/*.java javac server/*.java javac util/*.java 4.Type java server.Server	Console display: <pre>===== SERVER IS RUNNING..... PORT:1234 =====</pre>	Pass
TC02	Client connection	Verify that the client can connect successfully to the running server	1.Open the project in VS Code 2.Open terminal 3.Type java gui.LoginGUI 4.Enter the following information: Username: Giang Server IP: 127.0.0.1 Port: 1234 5.Click the Connect button	1.Client connects successfully to the server. 2.The Chat Room window is displayed. 3.The server console displays 4.The client receives the welcome message 5.The connection status is shown as Connected.	Pass
TC03	Login without username	Verify that the system prevents users from connecting when the username field is empty	1.Open the project in VS Code 2.Open terminal 3.Type java gui.LoginGUI 4.Leave username empty 5.Click the Connect button	1.The system displays an error message. 2.The client cannot connect to the server. 3.The Login window remains open.	Pass
TC04	Login with Invalid IP Address	Verify that the client cannot connect using an invalid server IP address.	1.Open the project in VS Code 2.Open terminal 3.Type java gui.LoginGUI 4.Type different IP 5.Click the Connect button	1.The system displays Connection Failed 2.Chat Room is not opened 3.The client remains on the Login window	Pass
TC05	Login with Invalid Port	Verify that the client cannot connect using an invalid port number	1.Open the project in VS Code 2.Open terminal 3.Type java gui.LoginGUI 4.Type different port 5.Click the Connect button	1.The connection fails 2.Chat Room is not displayed 3.An error message is shown	Pass
TC06	Login with Non-numeric Port	Verify that the system rejects a port value containing non-numeric characters	1.Open the project in VS Code 2.Open terminal 3.Type java gui.LoginGUI 4.Type letters in port instead of number 5.Click the Connect button	1.The system displays the message: "Port must be a number." 2.The client is not connected to the server 3.The Chat Room window is not opened	Pass
TC07	Broadcast Message	Verify that a public message is received by all connected users	1. Start the server. 2. Open two or more clients (e.g., Giang, Huy, Tuan). 3. Connect all clients to the server. 4. From client Giang, type: Hello	1. The message is sent successfully. 2. All connected clients receive and display the message. 3. The sender also sees the message in the chat room.	pass
TC08	Multiple Clients Join Chat Room	Verify that multiple users can connect simultaneously	1. Start the server. 2. Open three client windows. 3. Login with different usernames (e.g., Giang, Huy, Tuan). 4. Connect all clients to the server.	1. Each client connects successfully. 2. Each client enters the Chat Room. 3. The online user list is updated for every client. 4. The server console displays each successful connection.	pass

Proof

```
ges' '-cp' 'C:\Users\CHAU GIANG\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Chat-app_f5117b1\bin' 'server.Server'
=====
SERVER IS RUNNING...
Port: 1234
=====
```



TC09	Private Message	Verify that a private message is sent only to the selected recipient.	1. Start the server. 2. Open two client windows and connect with different usernames (tuan, giang and huy). 3. In dao's Chat Room, select user "huy" 4. Type: hello huy 5. Click the Send button.	1. The private message is sent successfully. 2. Only user "huy" receives the message. 3. Other connected users cannot see the message. 4. The sender (giang) sees the message displayed as: [PM][15:16:54] to huy: hello huy. 5. No error message is displayed.	pass
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